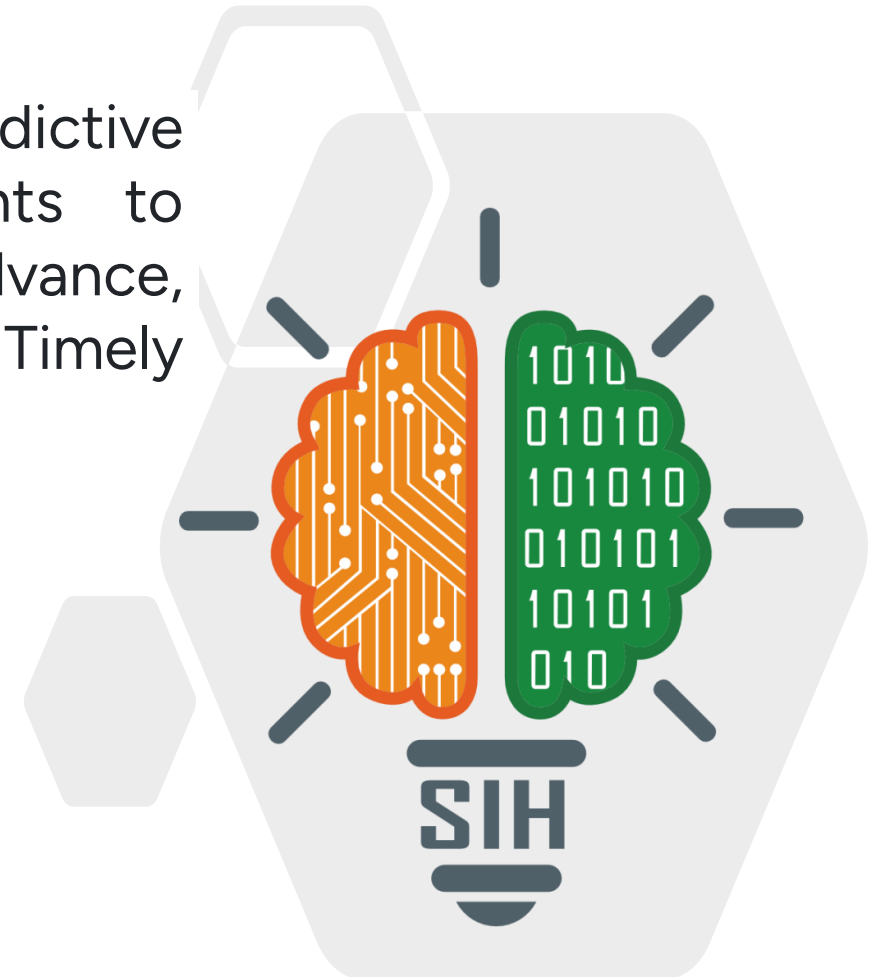


SMART INDIA HACKATHON 2025



- **Problem Statement ID – 25257**
- **Problem Statement Title** - Development of a Predictive Analytics Framework for Cybercrime Complaints to Forecast Likely Cash Withdrawal Locations in Advance, Enabling Generation of Actionable Intelligence for Timely and Proactive Cyber Crime Intervention.
- **Theme - BlockChain and CyberSecurity**
- **PS Category - Software**
- **Team ID - 70988**
- **Team Name - SKYVAR**



1. Our Proposed Solution : CIPHER



Hybrid AI Core for Prediction: Implement a Geospatial AI/ML Engine (BERT/XGBoost) that analyzes multilingual complaint text to predict future cash-out hotspots with a high-confidence risk score.



Zero-Latency Real-Time Alerting: Utilize WebSockets and Kafka to instantly push Critical Alerts to LEAs and FIs, eliminating manual delays critical for fraud mitigation.



Advanced Fraud Network Detection: Integrate a Neo4j Graph Database to perform deep linkage analysis, actively mapping and identifying complex, organized fraud rings across multiple accounts.



Automated Fund Freeze Integration: Deploy Smart Contracts (Hyperledger) to enable the instantaneous, non-repudiable freezing of funds via integrated Bank APIs, bypassing human bottlenecks.



Targeted LEA Resource Deployment: Translate AI predictions into an actionable Risk Heatmap that guides LEAs to pre-emptively deploy teams only to zones of maximum predicted risk.

2. Major Problems Addressed

01

Strategic Blind Spot: Lack of Predictive Intelligence 🎯

02

Operational Failure: The Velocity and Volume Crisis 📉

03

Fragmentation: Poor Inter-Agency Coordination 🌐

04

Limited Intervention Capabilities 🛑

3. Novelty and Innovation

1

Hybrid AI Core for Predictive Hotspots

2

Smart Contract Fund Freeze Automation

3

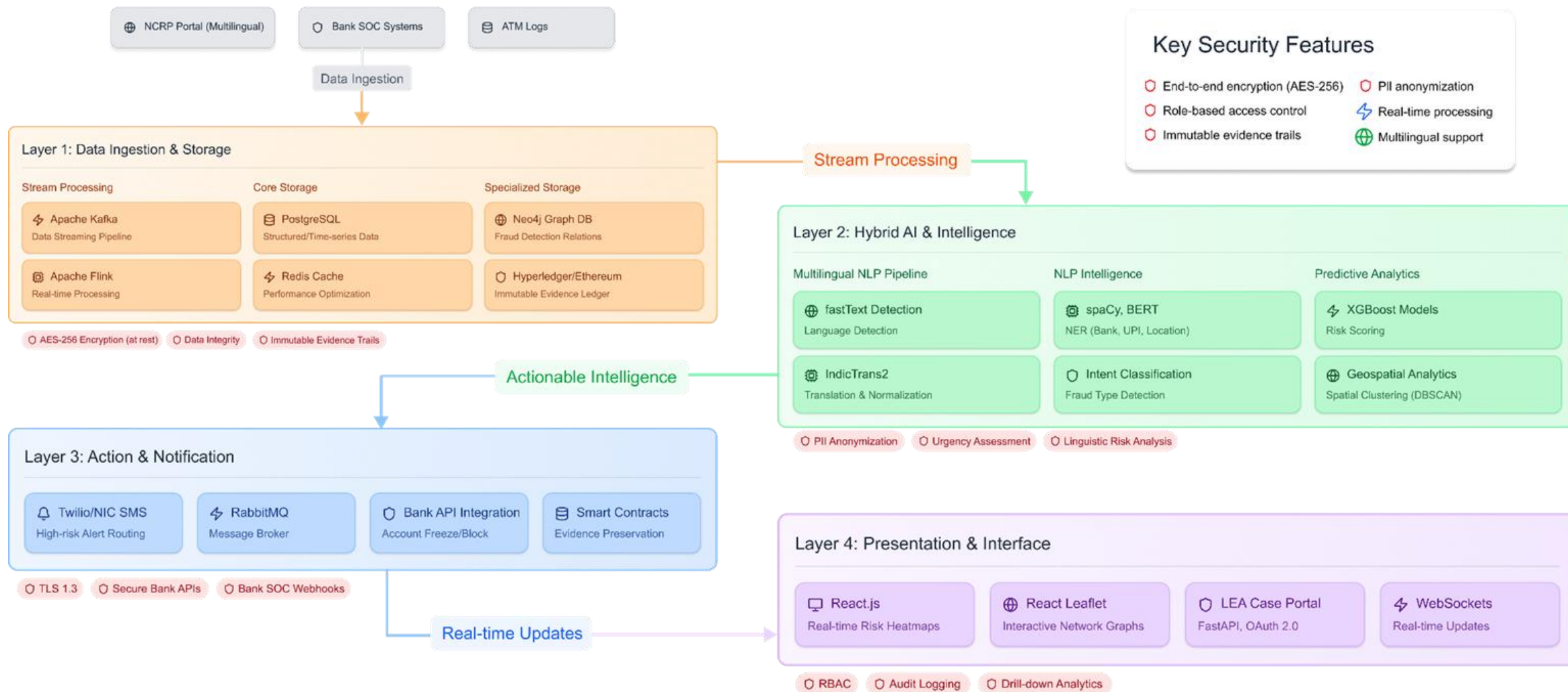
Real-Time, Multi-Channel Alerting

4

Advanced Fraud Ring De-Anonymization



[VIEW CODE ON
GITHUB](#)



FEASIBILITY AND VIABILITY

Multilingual Prediction
Solution : Utilize BERT/Indic NLP Models to standardize and extract context/intent from the translated text.

Trust & Auditability
Solution : Deploy a Hyperledger (Blockchain) Ledger to create a secure, non-repudiable audit trail for evidence.

Data Processing Speed

Solution : Implement Kafka for stream queuing and FastAPI Background Tasks for asynchronous processing.

Organized Crime

Solution : Integrate a Neo4j Graph Database to perform real-time network and linkage analysis.

Intervention Automation

Solution : Deploy Smart Contracts to automate the fund-freezing transaction via integrated Bank APIs.

Feasibility Challenges and its Viable Solutions

Novel Benefits :

01	Transformation to Proactive Policing	System shifts LEA focus from slow, post-incident logging to pre-emptive presence at predicted hotspots.
02	Maximize Fund Recovery Potential	Directly targets the failure point: enables intervention at the cash-out location, aiming to ↑ fund recovery rates significantly.
03	Geo-Spatial Resource Optimization	The Risk Heatmap allows LEAs to surgically deploy special teams, optimizing manpower against high-risk areas.
04	Mitigating Systemic Coordination Gaps	Establishes a Unified Defense framework, aligning Police, Banks, and I4C via real-time intelligence.
05	Reduction in Financial Loss Magnitude	Directly combats the massive scale of fraud: India lost ₹22,845 Crore in 2024; this system reduces that magnitude.

Possible Impacts :

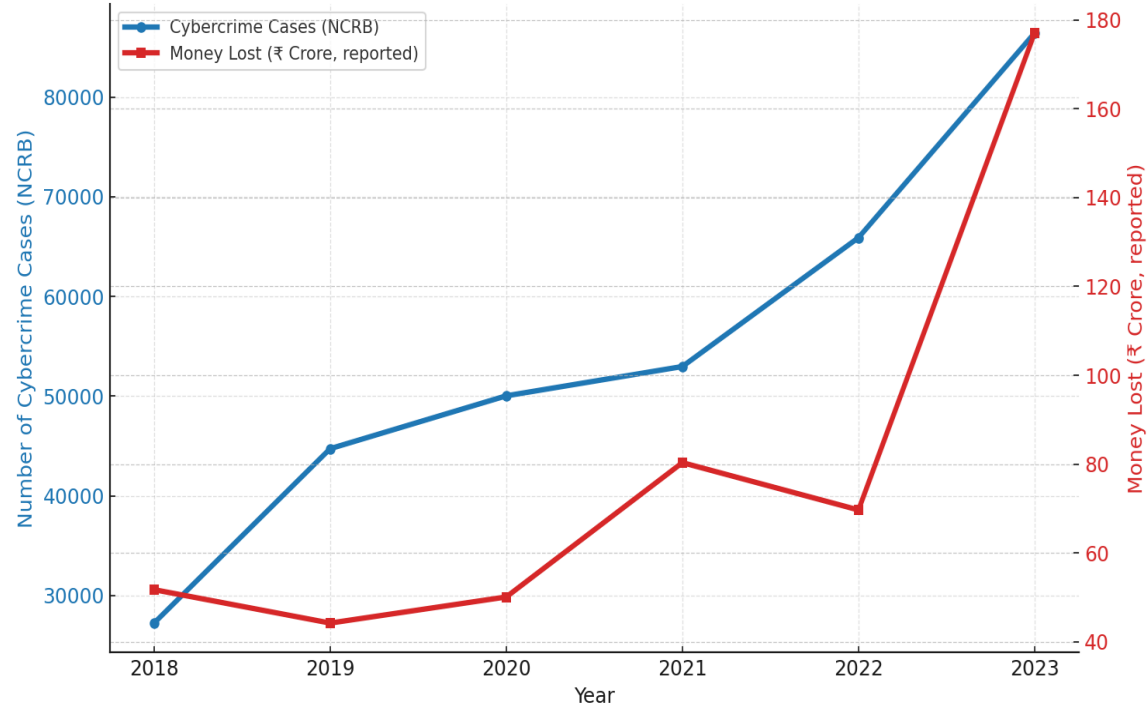
▲ Up to **98% Accuracy** in identifying fraud when using advanced AI/ML and anomaly detection systems.

▲ **15-25% reduction** in total losses by pre-emptively blocking high-value frauds.

▲ Reduction of **false positive rate by 50% to 60%**, ensuring a secure experience.

▲ **Place-based predictive policing** (your hotspot model) leads to a **20-35% faster response time** in target areas.

Need For CIPHER : Increasing Trend of CyberCrimes



Research Papers :

[1] A Graph Neural Network With Reinforcement Learning for Adaptive Financial Fraud Detection [↗](#)

[2] Crime Predictive Analysis Based on Geointelligence [↗](#)

[3] Integrating Smart Contracts with Machine Learning [↗](#)



[Youtube Link](#)